Def. Let $A \in M_{m\times n}(F)$ be given where $F = \mathbb{R}$ or $\mathbb{C}$. Define the supremum norm of A

$$\|A\|_p \overset{\text{def}}{=} \sup_{x\neq 0} \frac{\|Ax\|_p}{\|x\|_p} \overset{(*)}{=} \sup_{\|x\|_p=1} \|Ax\|_p$$

where $\|\cdot\|_p$ is the p-norm, that is,

$$\|\cdot\|_p : F^n \to \mathbb{R} : (a_i)_{i=1}^n \mapsto \left(\sum_{i=1}^n |a_i|^p\right)^{\frac{1}{p}}.$$

The equality $(*)$ is given by the fact that for any non-zero vector x, then

$$\frac{\|Ax\|_p}{\|x\|_p} = \left\|A \cdot \frac{x}{\|x\|_p}\right\|_p = \|Ax'\|_p, \text{ where } \|x'\| = 1.$$

In fact, the value $\|A\|_p$ is the smallest value C such that

$$\|Ax\|_p \leq C \cdot \|x\|_p \text{ for all non-zero } x \in F^n$$

That is, given $A \in M_{m\times n}(F)$ and $x \in F^n$,

$$\|Ax\|_p \leq \|A\|_p \|x\|_p$$

Prop. Given $A \in \mathcal{M}_{m\times n}(F)$ where $F = \mathbb{R}$ or $\mathbb{C}$,

$$\|A\|_1 = \max_{1\le j\le n} \sum_{i=1}^{m} |A_{ij}| \quad \text{and} \quad \|A\|_\infty = \max_{1\le i\le m} \sum_{j=1}^{n} |A_{ij}|$$

Proof. Given $x \in \mathbb{R}^n$ with $\|x\|_1 = 1$, denote $x = (x_1, \cdots, x_n)$.

$$\|Ax\|_1 \underbrace{=}_{\text{def}} \left\| \sum_{j=1}^{n} x_j \cdot Ae_j \right\|_1 \underbrace{\le}_{\text{Homo + Sub. add}} \sum_{j=1}^{n} |x_j| \cdot \|Ae_j\|_1 \le \sum_{j=1}^{n} |x_j| \cdot \max_{1\le j\le n} \|A\cdot e_j\|_1 = \max_{1\le j\le n} \|Ae_j\|_1.$$

And, take $x = e_k$ such that e_k maximizes the $\|Ae_j\|_1$, then we obtain the first.

Given $x \in \mathbb{R}^n$ with $\|x\|_\infty = 1$. Then, for any $1 \le k \le m$,

$$\left|(Ax)_k\right| = \left| \sum_{j=1}^{n} A_{kj} \cdot x_j \right| \le \sum_{j=1}^{n} |x_j| \cdot |A_{kj}| \le \sum_{j=1}^{n} |A_{kj}|$$

and the equality holds when $x = (1, 1, \cdots, 1)$. So proved. ☒

The $\|\cdot\|_2$ is so special, because it is induced by the inner product.
Note that the fact that: for any unitary $Q \in \mathcal{M}_{n\times n}(F)$, Q preserves the norm. That is,

$$\|Qx\|_2^2 = \langle Qx, Qx\rangle = \langle x, x\rangle = \|x\|_2^2$$

Thm. Given $A \in \mathcal{M}_{m\times n}(F)$ where $F = \mathbb{R}$ or $\mathbb{C}$,

$$\|A\|_2 = \sigma_{max}, \text{ the largest singular value of } A.$$

Proof. Given A, there is a Singular value decomposition,

$$A = U\Sigma V^* \text{ where } U \in \mathcal{M}_{m\times m},\ V \in \mathcal{M}_{n\times n} \text{ are unitary,}$$

and $\Sigma = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{pmatrix} \in \mathcal{M}_{m\times n}$, $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0$

So,

$$\|Ax\|_2 = \|U\Sigma V^* x\|_2 = \|\Sigma V^* x\|_2 = \|\Sigma y\|_2 \quad \text{where } y = V^* x$$

Since the unitary operator is isomorphism, and isometry, we can replace V^*x into y.
And, the p-Norm of diagonal matrix is the largest absolute value of its entry,
so proved.

Problems.

For $A=\begin{bmatrix}1 & -2 & 0\\ 0 & 1 & 2\end{bmatrix}$, Compute $\|A\|_1, \|A\|_\infty, \|A\|_F, \|A\|_2$.

Proof.

$\|A\|_1 = |-2|+|1|=3$, $\|A\|_\infty = |1|+|-2| = 3$.

$AA^* = \begin{bmatrix}5 & -2\\ -2 & 5\end{bmatrix}$

$\|A\|_F = \sqrt{1+4+1+4} = \sqrt{10}$. $\left(=\sqrt{tr(AA^*)}\right)$.

Meanwhile, $A^*A = \begin{bmatrix}1 & -2 & 0\\ -2 & 5 & 2\\ 0 & 2 & 4\end{bmatrix}$ has a characteristic polynomial $\chi_{A^*A} = t(t-3)(t-7)$

So, $\|A\|_2 = \sqrt{7}$.

Prove that Frobenius Cauchy-Schwarz inequality

$$|tr(A^T\cdot B)| \le \|A\|_F \cdot \|B\|_F.$$

And solve $\max_{\|X\|_F\le 1} tr(A^TX)$, find all maximizers for a non-zero A.

Rayleigh quotient.

Def. Let B be a n by n Hermitian matrix.
The Rayleigh quotient for $X \neq 0$ is

$$R(X) \overset{\text{def}}{=} \frac{\langle BX, X \rangle}{\|X\|^2}$$

Thm. Let B be a n by n Hermitian matrix.
Then, the spectrum $\sigma(B)$ is non-empty real numbers set and

$$\max_{X \neq 0} R(X) = \max \sigma(B) \quad \text{and} \quad \min_{X \neq 0} R(X) = \min \sigma(B).$$

Proof. Let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be an orthonomal basis for B such that each α_i is eigenvector of B corresponding to $\lambda_i \in \sigma(B)$.
Now, given $X \in F^n \setminus \{0\}$, $X = \sum_{i=1}^{n} a_i \cdot \alpha_i$ for some $a_i \in F$, so that

$$R(X) = \frac{\langle BX, X \rangle}{\|X\|^2} = \frac{\langle \sum a_i \cdot \lambda_i \cdot \alpha_i, \sum a_i \cdot \alpha_i \rangle}{\|X\|^2}$$

$$= \left(\sum_{i=1}^{n} \lambda_i \cdot |a_i|^2 \right) \Big/ \left(\sum_{i=1}^{n} |a_i|^2 \right)$$

$$\leq \max \sigma(B) \cdot \left(\sum_{i=1}^{n} |a_i|^2 \right) \Big/ \left(\sum_{i=1}^{n} |a_i|^2 \right),$$

So, the inequality is obtained, and take $X = \alpha$ such that $\alpha \in \mathcal{B}$ satisfying $B\alpha = \max \sigma(B) \alpha$, so we obtain $R(\alpha) = \max \sigma(\alpha)$.
Similarly, the minimum can be obtained. ∎

Lemma. Let A be an any square matrix over $\mathbb{C}$ or $\mathbb{R}$.
Then, λ is eigenvalue of $A^*A \Leftrightarrow \lambda$ is eigenvalue of AA^*.

Proof. Let $\lambda \in F$ be an eigenvalue of A^*A.
If $\lambda = 0$, then $\det(A^*A) = \det A^* \cdot \det A = \det(AA^*) = 0$, so
AA^* has 0 as an eigenvalue. And the converse is similarly shown.
If $\lambda \neq 0$, then for some non-zero vector X, $A^*AX = \lambda X$. Then,

$AA^*AX = A(\lambda X) = \lambda AX$. But, if $AX = 0$, then $A^*AX = 0 = \lambda X$,
So $AX \neq 0$,

Therefore, AA^* has λ as an eigenvalue. The converse can be shown,
exactly same way.

Corollary. Let $A \in M_{n\times n}(F)$ be an invertible matrix. Then,

$$\|A^{-1}\| = \frac{1}{\sigma_{min}(A)}$$

Proof. Let $\lambda_1 \geq \cdots \geq \lambda_n$ be eigenvalues of A^*A. By the lemma above,
these are also eigenvalues of AA^*. And, since

$$\|A^{-1}\|^2 = \text{the largest eigenvalue of } (A^{-1})^*(A^{-1}) = (AA^*)^{-1},$$

and $(AA^*)^{-1}$ has eigenvalues as

$$\lambda_n^{-1} \geq \cdots \geq \lambda_1^{-1},$$ (since $BX = \lambda X \Rightarrow \lambda^{-1} = B^{-1}X$, if $\det B \neq 0$)

So $\|A^{-1}\| = \sqrt{\lambda_n^{-1}} = 1/\sigma_{min}(A)$. ∎